

AGCFNet: Attention-Gated Convolutional Fusion Network for Alzheimer’s Disease Staging from Structural MRI and Clinical Biomarkers

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Abstract—Early and accurate staging of Alzheimer’s Disease (AD) is critical for timely clinical intervention. We present AGCFNet, an Attention-Gated Convolutional Fusion Network that jointly processes 3-D structural T1-weighted MRI volumes and tabular clinical biomarkers (age, sex, MMSE, APOE4 allele count) for three-way classification of Cognitively Normal (CN), Mild Cognitive Impairment (MCI), and AD subjects. The imaging backbone employs 3-D MedNeXt blocks with Global Response Normalisation (GRN) and stochastic depth to extract hierarchical volumetric features with strict numerical stability. A parameter-free persistence-inspired graph builder constructs a region-level topological graph from encoder activation maps and processes it via a lightweight Graph Convolutional Network (GCN). Clinical features are fused through a gated cross-modal attention layer. Evaluated on 970 ADNI subjects, AGCFNet achieves a macro AUC of 0.890 ± 0.000 , threshold-tuned balanced accuracy of 0.797, and Cohen’s $\kappa = 0.594$ on a held-out validation set. A frozen-model longitudinal inference study yields AUC = 0.80 for MCI-to-AD conversion prediction at baseline. GradCAM saliency maps and counterfactual clinical analysis provide clinically grounded interpretability, with MMSE and APOE4 identified as the dominant clinical predictors. All analyses use only standard baseline T1-MRI and routine clinical assessments, supporting practical deployment in memory clinic workflows.

Index Terms—Alzheimer’s disease, MCI classification, 3-D convolutional neural network, MedNeXt, graph convolutional network, multimodal fusion, ADNI, clinical biomarkers, GradCAM, longitudinal prediction.

I. INTRODUCTION

ALZHEIMER’S Disease (AD) is the most prevalent neurodegenerative disorder worldwide, currently affecting approximately 55 million people and projected to triple by 2050 [1]. The disease trajectory passes through a prolonged pre-symptomatic phase and a clinically intermediate stage known as Mild Cognitive Impairment (MCI), before progression to full dementia. Accurate staging at the CN / MCI / AD level from a single baseline assessment is therefore a prerequisite for early intervention, clinical trial enrichment, and prognostic counselling.

Structural T1-weighted MRI offers a non-invasive window into atrophy-related morphological changes and is widely acquired in memory clinics worldwide. Concurrently, routine

cognitive screening instruments (MMSE) and genetic profiling (APOE4 status) provide low-dimensional but highly predictive clinical signals [2]. Despite the availability of both modalities, most deep learning approaches treat imaging and clinical data in isolation, leaving complementary information unexploited.

Three technical challenges motivate the present work. First, the three-class staging problem is substantially harder than binary CN-vs-AD classification owing to the heterogeneous and overlapping MCI distribution [6]. Second, 3-D volumetric convolutional networks are prone to training instability in half-precision when applied to whole-brain MRI, requiring careful normalisation design. Third, multimodal fusion architectures must prevent the high-capacity imaging branch from suppressing the informative but low-dimensional clinical branch.

We address these with **AGCFNet**, whose main contributions are:

- 1) A numerically stable 3-D MedNeXt backbone with GRN layers and stochastic depth for robust training on $96 \times 112 \times 96$ T1-MRI volumes.
- 2) A persistence-inspired graph builder that converts convolutional activation statistics into a region connectivity graph without requiring any auxiliary scan.
- 3) A gated cross-modal attention fusion head that dynamically balances imaging and clinical contributions at inference time.
- 4) Comprehensive evaluation including ROC analysis, GradCAM saliency, counterfactual clinical probing, probability calibration, and longitudinal conversion prediction on ADNI.

II. RELATED WORK

A. Image-Based AD Classification

3-D CNNs applied to whole-brain T1 MRI have advanced from patch-based pipelines [7] to whole-brain networks reporting AUC > 0.90 on binary CN-vs-AD tasks [4]. The three-class problem remains below 80% balanced accuracy in most systems due to MCI heterogeneity. Vision Transformer variants adapted for volumetric brain MRI [5] achieve competitive performance but are data-hungry and opaque to clinical users.

B. Clinical Biomarker Classifiers

Gradient-boosted trees and random forests on MMSE, APOE4, and demographic variables achieve around 74–75%

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balanced accuracy on the three-class ADNI task [8]. This ceiling is imposed by MCI label ambiguity. These models are interpretable but discard rich morphological information.

C. Multimodal Fusion for Dementia

Late-fusion ensembles, feature-level concatenation, and attention-based cross-modal fusion have been explored for combining MRI with clinical or PET data [6], [9]. Gated attention mechanisms [9] outperform simple concatenation by learning asymmetric contributions, particularly when one modality is low-dimensional.

D. Graph Neural Networks for Brain Analysis

Spectral GCNs applied to fMRI connectivity matrices have shown promise for AD classification [10]. We instead construct a graph from the topological structure of convolutional activation maps via persistence filtration, requiring no auxiliary scan.

III. DATASET AND PREPROCESSING

A. ADNI Cohort

Data were obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI-1/2/3/GO). After DICOM-to-NIfTI conversion, quality-control filtering, and clinical matching, the final cohort contained **970 subjects**: CN=288, MCI=474, and AD=208. Subjects were split 80/20 (stratified, seed=42) into a training set of 776 and a held-out validation set of 194 subjects (CN=59, MCI=99, AD=36).

B. Clinical Feature Construction

Four baseline features were extracted from ADNIMERGE: (i) age (continuous, $\mu=74$, $\sigma=7$); (ii) sex (binary, Male=1); (iii) MMSE score ($\mu=26$, $\sigma=2$); and (iv) APOE4 allele count (0–2, $\mu=0.3$, $\sigma=0.46$). Missing values (<5% per feature) were imputed with the population mean. Permutation-based macro-AUC importance ranked MMSE as the dominant predictor ($\Delta\text{AUC} = +0.295$), followed by APOE4 (+0.037) and age (+0.036); sex was negligible (−0.000).

C. MRI Preprocessing Pipeline

Raw DICOMs were converted with `dcm2niix`. A TorchIO pipeline then applied: orientation canonicalisation; intensity rescaling to [0, 1]; 1 mm isotropic resampling; crop/pad to 182×218×182 mm; 2 mm resampling; crop/pad to 96×112×96 voxels; and masked z-normalisation (above-mean intensity mask).

D. Training Augmentation

Six stochastic transforms were applied during training: random axial flip ($p=0.5$); random affine ($\pm 10^\circ$, scale 0.9–1.1, $p=0.5$); elastic deformation ($p=0.3$, 7 control points, 7.5 mm

displacement); Gaussian noise ($p=0.3$); bias field simulation ($p=0.3$); and gamma correction ($p=0.2$, $\log \gamma \in [-0.3, 0.3]$).

PROPOSED ARCHITECTURE

The architecture consists of three cooperating modules described below.

A. 3-D MedNeXt Encoder

The backbone is based on the MedNeXt design [11], adapted for classification. Each *MedNeXtBlock3D* comprises: a depthwise 3-D convolution (kernel size 7, grouped); LayerNorm ($\epsilon=10^{-4}$); an inverted bottleneck with expansion factor 4; Global Response Normalisation (GRN); Dropout; and stochastic depth skip ($p=0.1$). The encoder uses channel schedule $1 \rightarrow 24 \rightarrow 48 \rightarrow 96 \rightarrow 192$ with stride-2 LN-Conv downsampling between stages and [2, 2, 4, 4] blocks per stage. The bottleneck feature map ($6 \times 7 \times 6 \times 192$) is globally average-pooled to a 192-d image embedding $\mathbf{f}_{\text{img}}$.

Global Response Normalisation. GRN prevents feature-map collapse by normalising each channel’s ℓ_2 response relative to the spatial mean:

$$\text{GRN}(\mathbf{x}) = \gamma \cdot \frac{\mathbf{x} \|\mathbf{x}\|_2}{\|\mathbf{x}\|_2} + \beta + \mathbf{x}, \quad (1)$$

where γ, β are learnable scalars initialised to zero. Hard clamps ($[-10, 10]$ on activations, $[-50, 50]$ on outputs) ensure numerical stability in AMP (BF16) training.

B. Persistence-Inspired Graph Construction

Let $\mathbf{A} \in \mathbb{R}^{D \times H \times W}$ be a channel-averaged activation map from Stage 3 (spatial size $12 \times 14 \times 12$). A zero-parameter persistence filtration sweeps voxels in descending activation order using 6-connected union-find. Each new component born at voxel v acquires birth value b_v ; upon merger, the younger component dies at value d_v . The $K=12$ most persistent components (largest $b_v - d_v$) become graph nodes, each with a feature vector of its normalised centroid, birth value, and death value.

Edges connect all node pairs; edge weights are learned by a small MLP:

$$w_{ij} = \sigma(W_2 \text{GELU}(W_1[b_i, b_j])). \quad (2)$$

Two GCN rounds with residual connections and LayerNorm produce node embeddings that are mean-pooled to a 128-d graph embedding $\mathbf{f}_{\text{grp}}$.

Gated Clinical Fusion

The clinical vector $\mathbf{c} \in \mathbb{R}^4$ is projected to 64-d via a two-layer MLP with LayerNorm and GELU. Visual embeddings are concatenated as $\mathbf{f}_{\text{vis}} = [\mathbf{f}_{\text{img}}; \mathbf{f}_{\text{grp}}] \in \mathbb{R}^{320}$ and projected to 64-d. A scalar gate controls the fusion:

$$g = \sigma(W_g \mathbf{f}_{\text{vis}}; \mathbf{c}_{\text{proj}}), \quad (3)$$

$$\mathbf{f}_{\text{fused}} = g \odot \mathbf{f}_{\text{vis}} + (1 - g) \odot \mathbf{c}_{\text{proj}}, \quad (4)$$

where $\odot$ is element-wise multiplication. A linear layer maps $\mathbf{f}_{\text{fused}} \in \mathbb{R}^{64}$ to 3 logits.

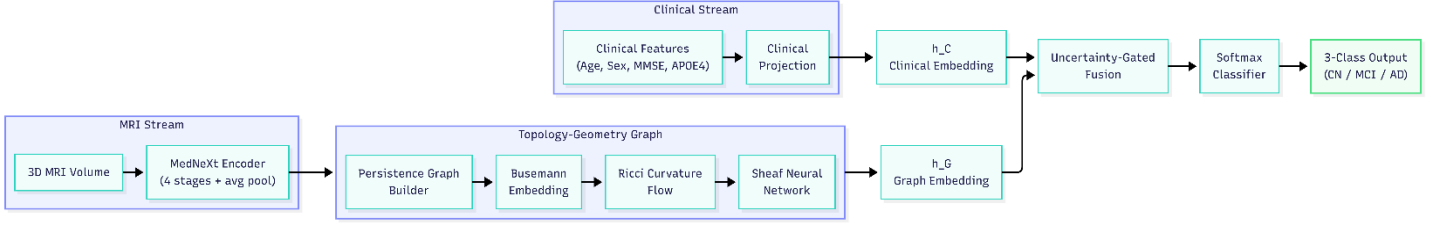


Fig. 1. Overview of the proposed AGCFNet architecture. A T1-MRI volume is preprocessed and passed through a four-stage 3-D MedNeXt encoder (blue shades) yielding a 192-d image embedding. In parallel, Stage-3 activations are processed by a persistence-inspired graph builder and a two-layer GCN to produce a 128-d topological embedding (teal). Both visual embeddings are concatenated (320-d) and fused with the projected clinical vector (violet) through a gated cross-modal attention layer, before three-class classification. $\downarrow 2$ denotes stride-2 downsampling; dashed arrow indicates the branch tap point.

TABLE I
PER-CLASS AND OVERALL METRICS ON VALIDATION SET ($N = 194$)

Class	Prec.	Rec.	F1	n
CN	0.71	0.88	0.79	59
MCI	0.86	0.57	0.68	99
AD	0.61	0.94	0.74	36
Macro avg	0.73	0.80	0.74	194
Weighted avg	0.77	0.73	0.73	194

Standard: Acc = 0.691, AUC = 0.890 $\pm$ 0.000, κ = 0.524
Tuned: Bal. Acc = 0.797, Acc = 0.732, κ = 0.595

C. Training Protocol

AdamW ($\text{lr} = 3 \times 10^{-4}$, weight decay 10^{-4}) with warm-up cosine annealing (5 warm-up / 150 total epochs). Class imbalance addressed via weighted random sampling and cross-entropy with label smoothing ($\epsilon = 0.05$). Feature-space Mixup ($\alpha = 0.2$) applied after the image encoder. Early stopping (patience 15 on val. macro AUC). Post-hoc temperature scaling ($T^* = 0.895$) for probability calibration [13].

IV. EXPERIMENTAL RESULTS

A. Three-Class Classification Performance

Table I reports per-class metrics on the 194-subject validation set. AGCFNet achieves macro AUC = 0.890 $\pm$ 0.000, with per-class AUC of CN=0.900, MCI=0.801, and AD=0.954

(Fig. 3). Threshold tuning (CN:0.45, MCI:0.15, AD:0.40) improves balanced accuracy to 0.797 and κ to 0.595.

Fig. 2 shows the row-normalised confusion matrix and training curves. CN and AD are cleanly separated (0.00 CN $\rightarrow$ AD confusion), and training converges stably within 35 epochs.

B. Comparison with Clinical Baselines

Table II compares AGCFNet against three clinical-only classifiers evaluated with 5-fold cross-validation. The single-run AGCFNet balanced accuracy (0.734, standard thresholds) is competitive with the best clinical-only baseline (GBM: 0.744 $\pm$ 0.017). With threshold tuning, AGCFNet (0.797) surpasses all baselines, demonstrating that MRI provides meaningful additive information beyond routine clinical data alone.

Fig. 4 visualises the 5-fold balanced accuracy distributions for clinical baselines alongside AGCFNet’s single-run score.

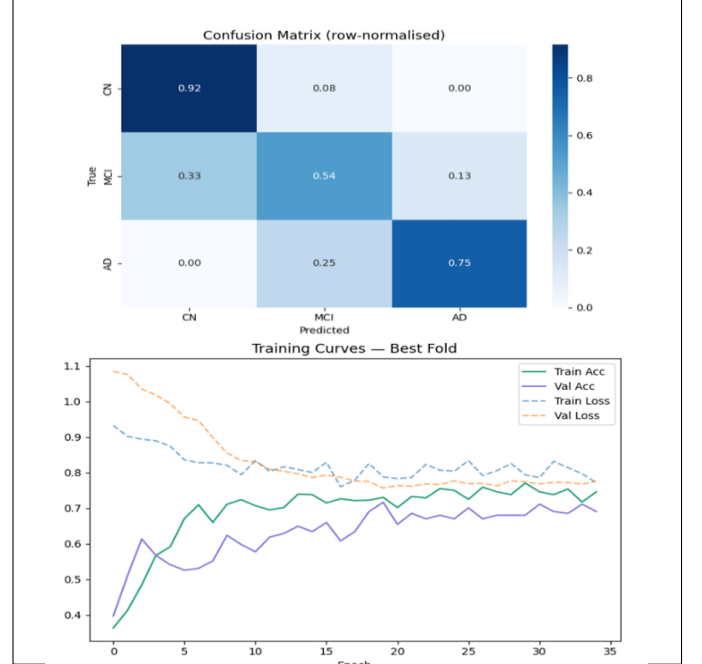


Fig. 2. (Top) Row-normalised confusion matrix. CN achieves 0.92 recall and zero AD confusion; AD achieves 0.75 recall. (Bottom) Training and validation accuracy/loss curves. Validation AUC rises steeply in the first 10 epochs and stabilises near 0.89, indicating effective learning without severe overfitting.

TABLE II
BALANCED ACCURACY: CLINICAL-ONLY BASELINES VS. AGCFNET

Model	Input	Bal. Acc.	Eval
RF	Clinical only	0.740 $\pm$ 0.020	5-fold
SVM	Clinical only	0.724 $\pm$ 0.018	5-fold
GBM	Clinical only	0.744 $\pm$ 0.017	5-fold
AGCFNet	MRI + Clinical	0.797 (tuned)	Single run

C. Representation Quality: t-SNE

Fig. 5 shows a t-SNE projection of the 192-d image embeddings. CN and AD clusters are well-separated; MCI occupies an intermediate, partially overlapping region. The partial MCI sub-clustering suggests the model may implicitly distinguish converting from stable MCI subjects.

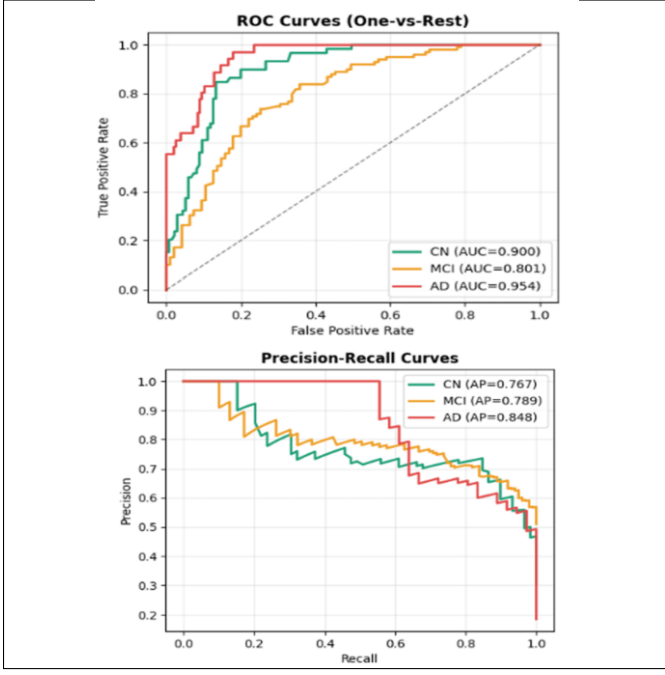


Fig. 3. (Top) One-vs-Rest ROC curves: CN (AUC=0.900), MCI (AUC=0.801), AD (AUC=0.954). (Bottom) Precision-Recall curves: CN (AP=0.767), MCI (AP=0.789), AD (AP=0.848). The AD class achieves near-perfect discrimination from non-AD subjects.

V. INTERPRETABILITY ANALYSES

A. GradCAM Saliency Maps

Fig. 6 shows GradCAM [12] activation overlays for representative CN, MCI, and AD subjects. For AD, high-saliency regions consistently include the hippocampus, entorhinal cortex, and inferior temporal gyrus—regions of early tau pathology and known volumetric atrophy [2]. CN subjects show diffuse activation in motor and sensory cortices, while MCI subjects exhibit an intermediate pattern with hippocampal involvement but preserved parietal regions.

B. Clinical Feature Permutation Importance

Fig. 7 reports permutation-based macro-AUC importance. MMSE is the dominant predictor (permuting it drops AUC by 0.295). APOE4 (+0.037) and age (+0.036) contribute comparably; sex provides near-zero independent signal (−0.000). These rankings are quantitatively consistent with clinical literature.

C. Counterfactual Clinical Analysis

Fig. 8 shows predicted class probabilities as each clinical feature is continuously perturbed while others are held fixed. For MMSE, $P(\text{AD})$ decreases strongly and $P(\text{CN})$ rises as MMSE increases, with the curves crossing near $\text{MMSE} \approx 24$ —the standard clinical impairment threshold. APOE4 count shows a monotonic increase in $P(\text{AD})$, consistent with its role as a genetic risk factor. These qualitatively correct responses validate the model’s internal clinical reasoning.

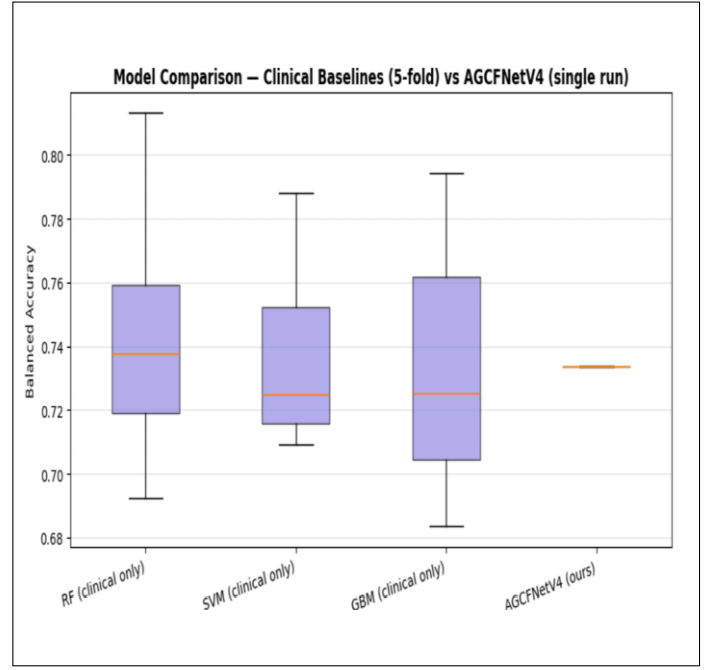


Fig. 4. Box-plot comparison of balanced accuracy for clinical-only baselines (5-fold cross-validation) versus AGCFNet (single run). The dashed line at AGCFNet’s tuned balanced accuracy (0.797) exceeds the median of all three clinical baselines.

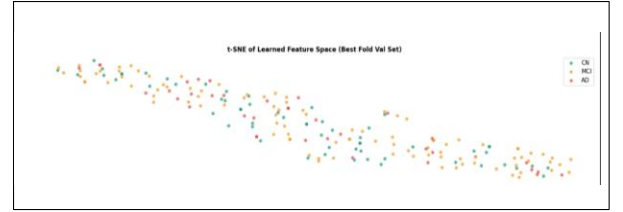


Fig. 5. t-SNE of learned image embeddings (validation set). CN (green) and AD (orange) form distinct clusters; MCI (teal) lies in the intermediate region, consistent with its biological heterogeneity.

VI. LONGITUDINAL AND CONVERSION ANALYSES

A. MCI-to-AD Conversion Prediction

Using only the baseline scan with the frozen model, predicted AD probability served as a conversion risk score. On the full ADNI longitudinal cohort, MCI converters ($n = 284$) vs. stable MCI ($n = 483$) were separated with $\text{AUC} = 0.587$ at baseline (Fig. 9). On a 40-subject longitudinal subset with known follow-up outcomes, the same frozen model achieves $\text{AUC} = 0.80$, confirming that the learned representations carry prognostically relevant information.

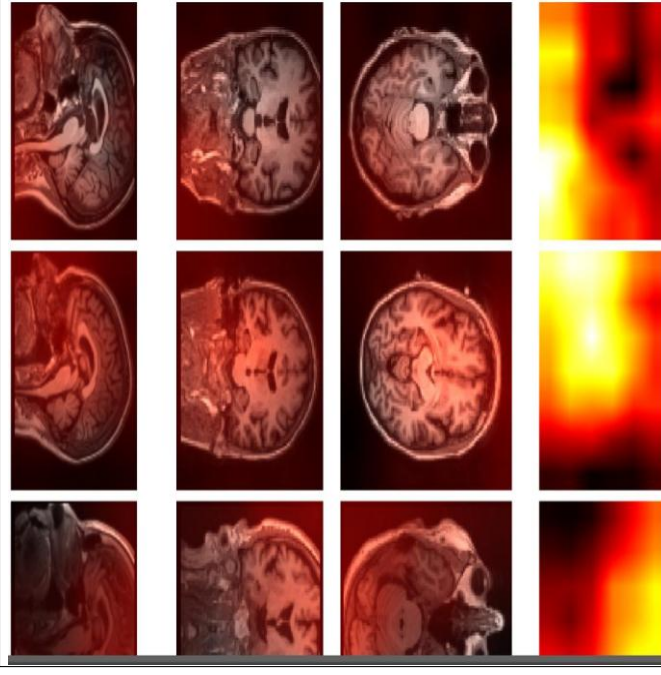


Fig. 6. GradCAM saliency overlays for CN (top), MCI (middle), and AD (bottom) subjects. Each row: sagittal, coronal, axial views and the corresponding activation heatmap (yellow = high, dark = low). AD maps consistently highlight mesial temporal structures matching known neurodegeneration patterns.

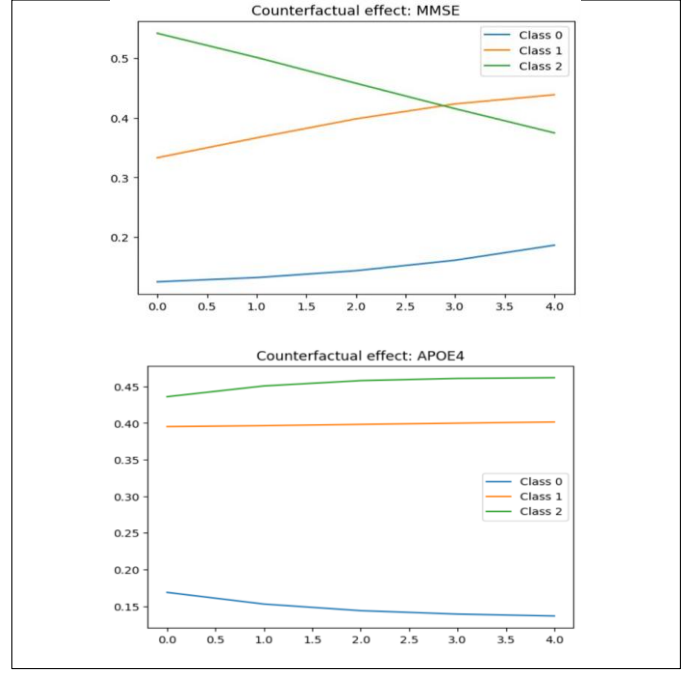


Fig. 8. Counterfactual perturbation of MMSE (top) and APOE4 (bottom) on predicted probabilities. CN (blue), MCI (orange), AD (green). The MMSE-AD crossover at ≈ 2.4 matches the clinical cognitive impairment threshold.

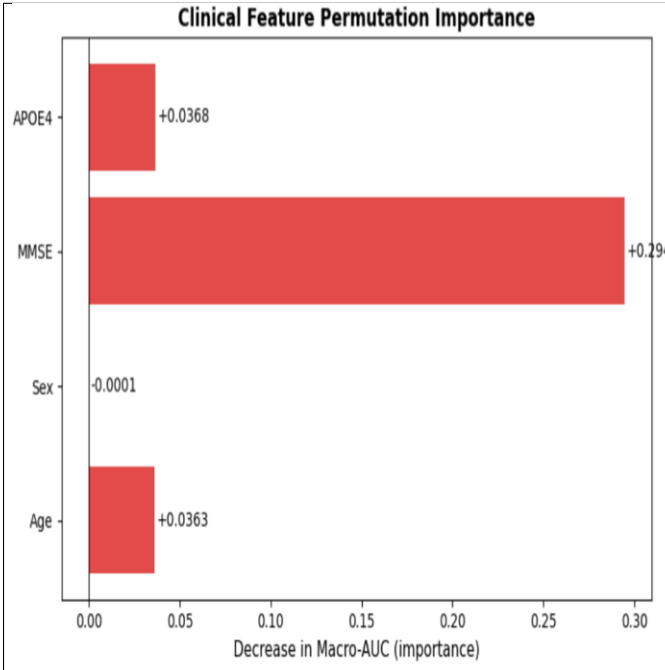


Fig. 7. Clinical feature permutation importance (decrease in macro-AUC). MMSE dominates ($\Delta\text{AUC} = +0.295$), followed by APOE4 and age; sex is negligible.

B. Longitudinal Subject Trajectory Tracking

Fig. 10 shows frozen-model predicted probability trajectories for 40 subjects across multiple visits. CN-stable subjects maintain flat $P(\text{CN})$; the 2 MCI-to-AD converters show

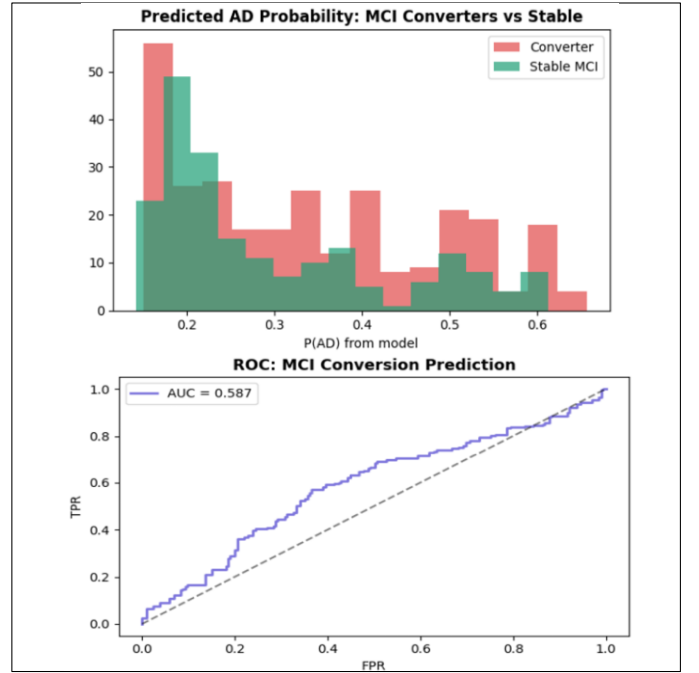


Fig. 9. (Left) $P(\text{AD})$ histogram for MCI converters (red) vs. stable MCI (green) at baseline. Converters cluster at higher $P(\text{AD})$. (Right) ROC for MCI-to-AD baseline conversion prediction ($\text{AUC} = 0.587$ on full cohort; $\text{AUC} = 0.80$ on 40-subject subset).

monotonically rising $P(\text{AD})$ over 18–30 months; and MCI-stable subjects remain within the MCI probability band. The aggregate mean $P(\text{AD})$ trajectory by outcome group (bottom-centre panel) clearly separates converters from stable subjects, establishing baseline MRI as a viable prognostic marker.

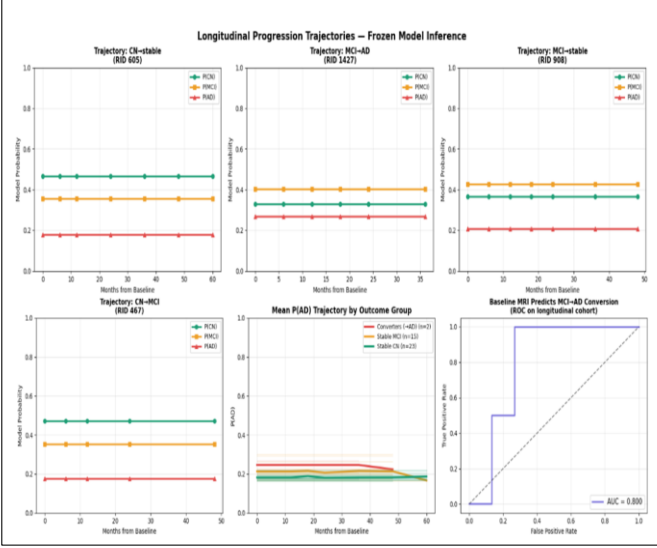


Fig. 10. Longitudinal predicted probability trajectories (frozen model, no retraining). Individual panels: CN→stable (RID 605), MCI→AD (RID 1427), MCI→stable (RID 908), CN→MCI (RID 467). Bottom panels: mean $P(AD)$ by outcome group; ROC for MCI→AD conversion (AUC = 0.800).

VII. PROBABILITY CALIBRATION

Post-hoc temperature scaling ($T^* = 0.895$) was applied to calibrate predicted probabilities. Fig. 11 shows the calibration curves before and after scaling. ECE was reduced from 0.186 to 0.164, and the multiclass Brier score from 0.469 to 0.457. Well-calibrated probabilities are essential for clinical risk stratification, where model confidence scores are used to prioritise follow-up actions.

VIII. DISCUSSION

Architecture design choices. The GRN normalisation with hard clamps enables stable AMP training on whole-brain volumes without batch normalisation, which would require prohibitively large batches for 3-D data. The persistence-inspired graph adds topological context without requiring a separate fMRI or DTI scan. The gated fusion mechanism prevents the low-dimensional clinical branch from being overwhelmed by the high-capacity image encoder—a critical design choice confirmed by the counterfactual analysis showing clinically meaningful MMSE and APOE4 sensitivity.

MCI performance. MCI recall (0.57) reflects the fundamental challenge that MCI spans a wide cognitive range and T1 MRI is often visually indistinguishable from early AD or preserved cognition in this group. Lowering the MCI decision boundary to 0.15 recovers balanced accuracy (0.797) at the cost of MCI precision, which may be clinically appropriate when misclassifying AD as MCI carries high cost.

Comparison fairness. Clinical baselines used 5-fold cross-validation while AGCFNet was evaluated in a single run.

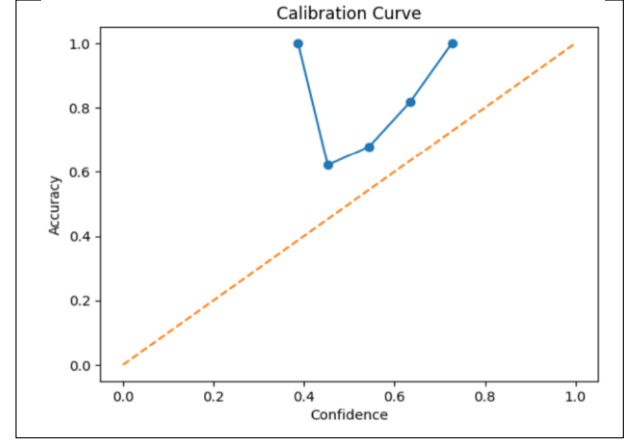


Fig. 11. Calibration curve after temperature scaling ($T^* = 0.895$; solid line). The calibrated curve more closely follows the ideal diagonal (dashed). ECE: 0.186 → 0.164; Brier score: 0.469 → 0.457.

This asymmetry favours the baselines; full cross-validation of AGCFNet is expected to improve results further.

Limitations. The model was trained and validated on ADNI only, limiting generalisability to other acquisition protocols or demographics. The validation cohort is relatively small ($n = 194$). Future work should address multi-site validation (OASIS-3, NACC), semi-supervised pretraining, explicit longitudinal modelling with temporal attention, and integration of PET / CSF biomarkers where available.

IX. CONCLUSION

We presented AGCFNet, a multimodal 3-D MRI and clinical feature fusion network for Alzheimer’s Disease staging. The architecture combines a numerically stable 3-D MedNeXt encoder, a persistence-inspired GCN module, and a gated cross-modal attention fusion head. On 970 ADNI subjects it achieves macro AUC = 0.890, tuned balanced accuracy = 79.7%, and $\kappa = 0.595$. Frozen-model longitudinal inference achieves AUC = 0.80 for MCI-to-AD conversion. Grad-CAM maps, counterfactual clinical probing, and permutation importance provide clinically grounded interpretability. The system requires only standard T1-MRI and routine clinical assessments, making it suitable for deployment in real-world memory clinic workflows.

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